

Customer Insight & Churn Predictions

Presentation By
Luche Louw

Data Science Practitioner Sprinter

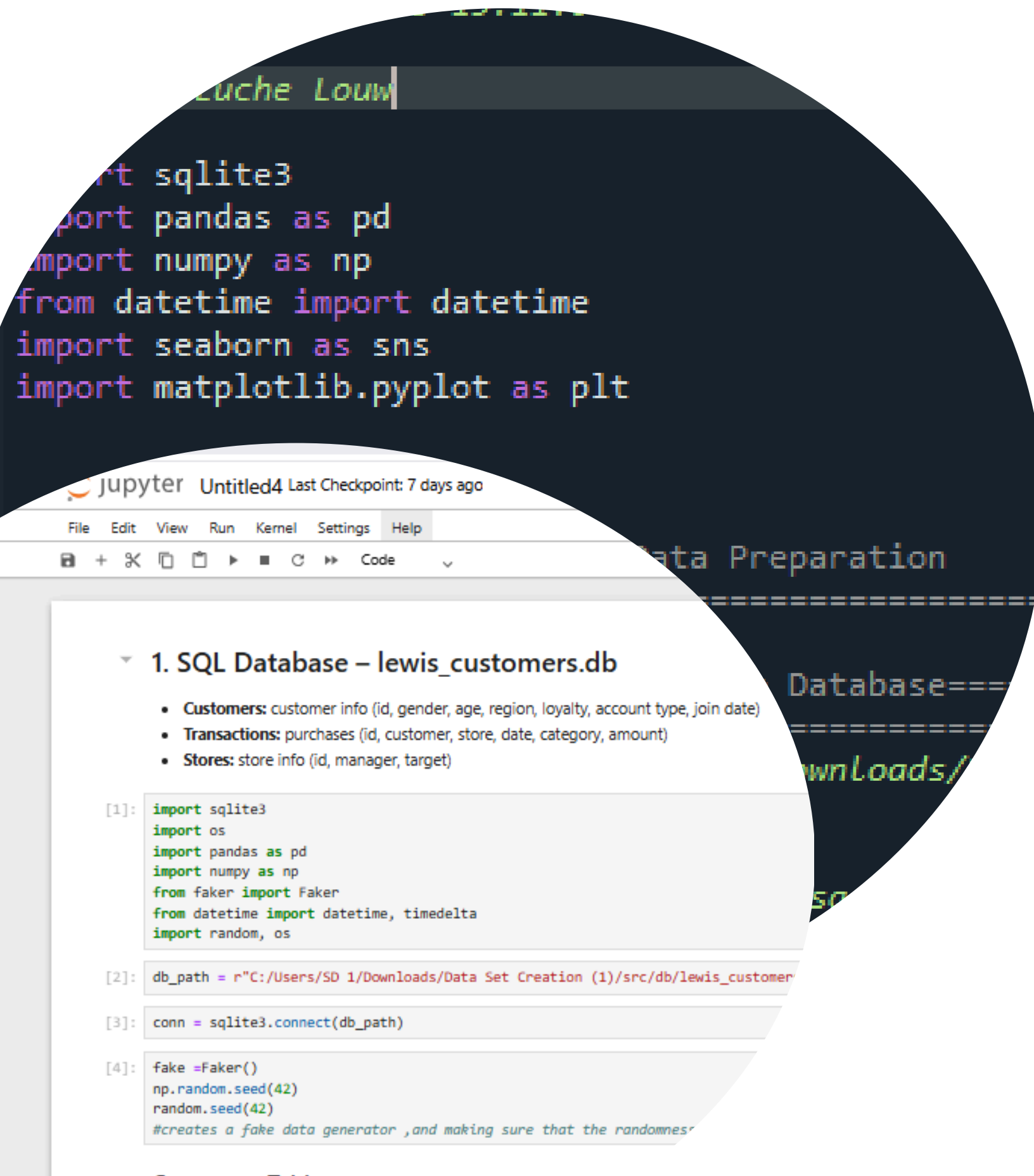


Objectives

The aim was to gather customer and purchase data and explore it in order to understand customer behaviour.

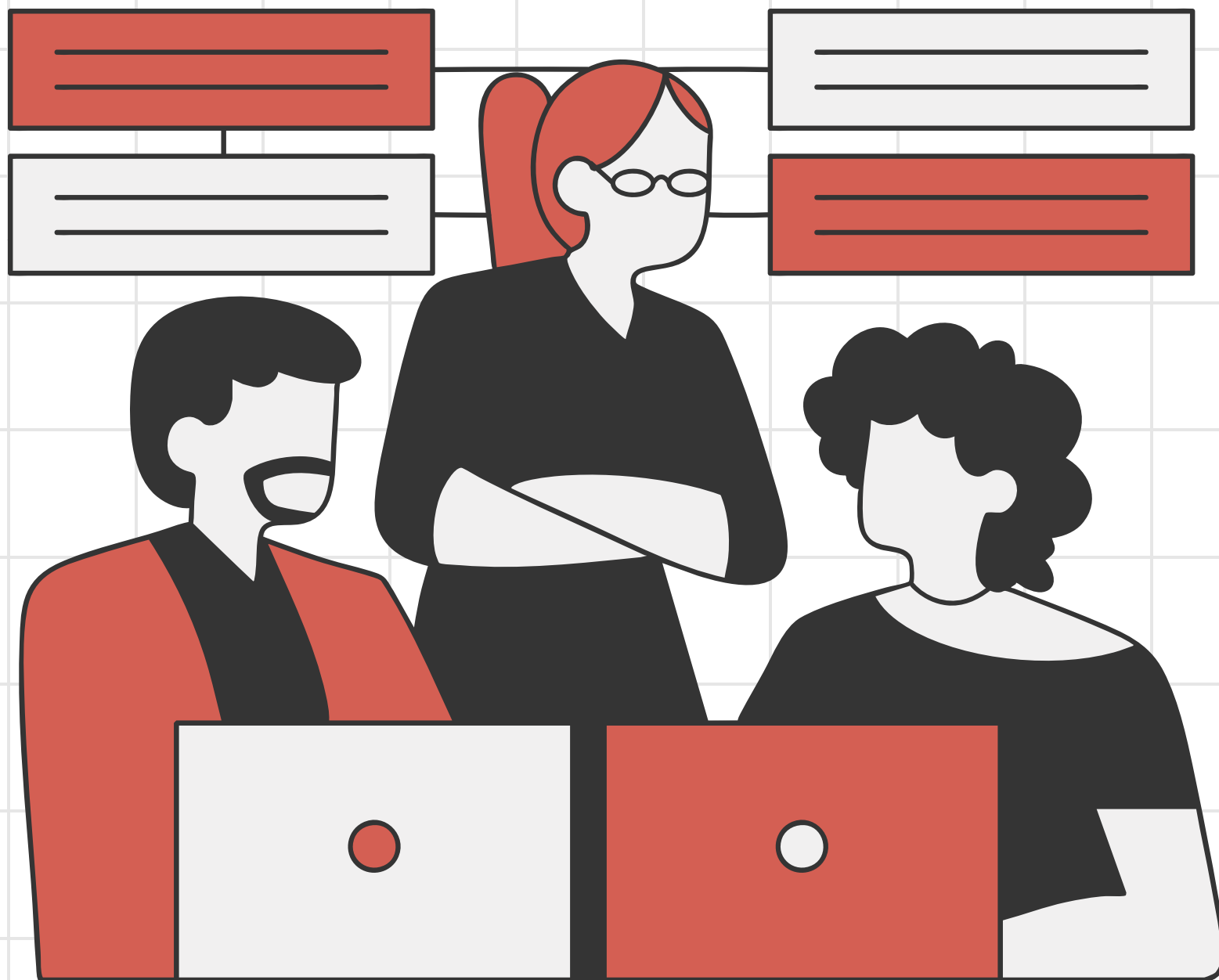
Build a predictive model to flag customers that are likely to churn.

The tools used to make this possible is Spyder IDE , Jupyter Notebook and most importantly PYTHON. Python enabled the usage of libraries such as SQLite3 ,Pandas , Scikit-Learn, Seaborn and Matplotlib.



Simulated Dataset Overview

A simulated database was used with artificially generated data, containing records for 200 customers over the span of one year.



A database called **Lewis.db** was created , which includes 3 tables:

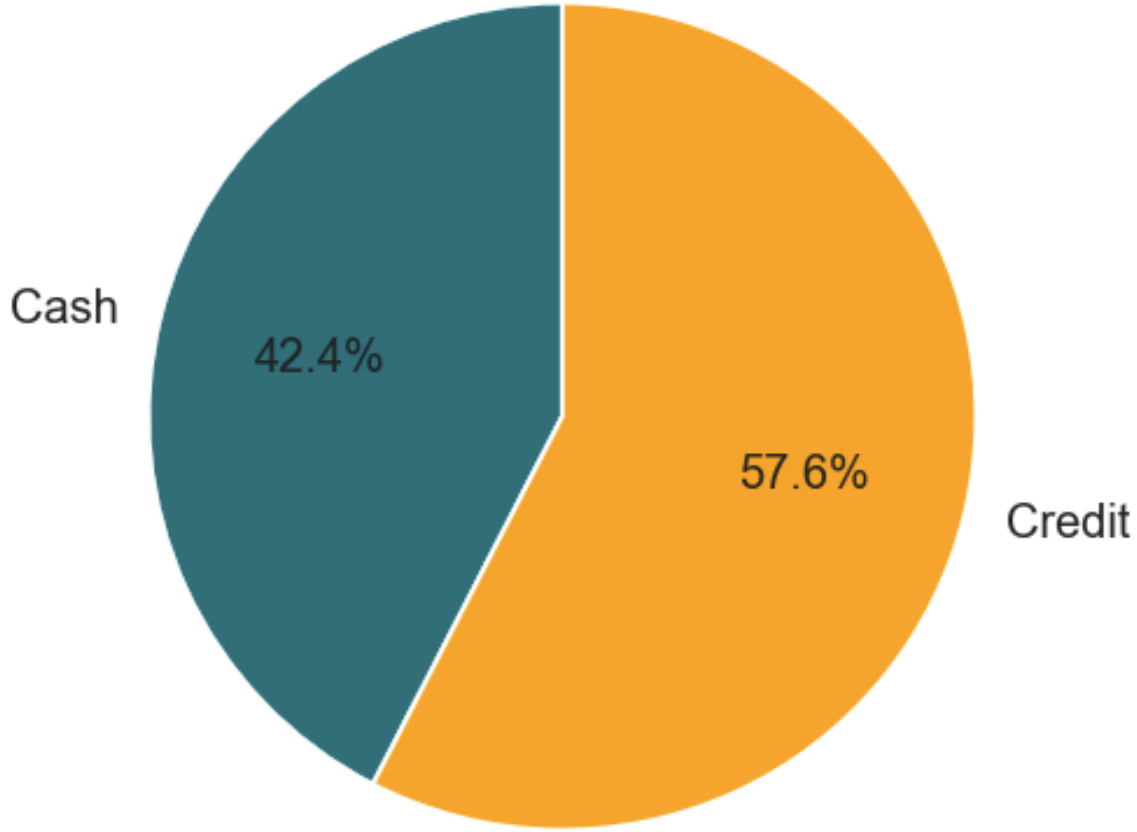
- **Stores**
- **Transactions**
- **Customers**

The tables were merged with a CSV file that contained customer **feedback information**. The purpose of the combined dataset is to:

- Reveal reasons for customer Churn
- Improve the accuracy of churn prediction models
- Classify types of churn risks and patterns
- Enable proactive customer retention actions

Customer Satisfaction Insights

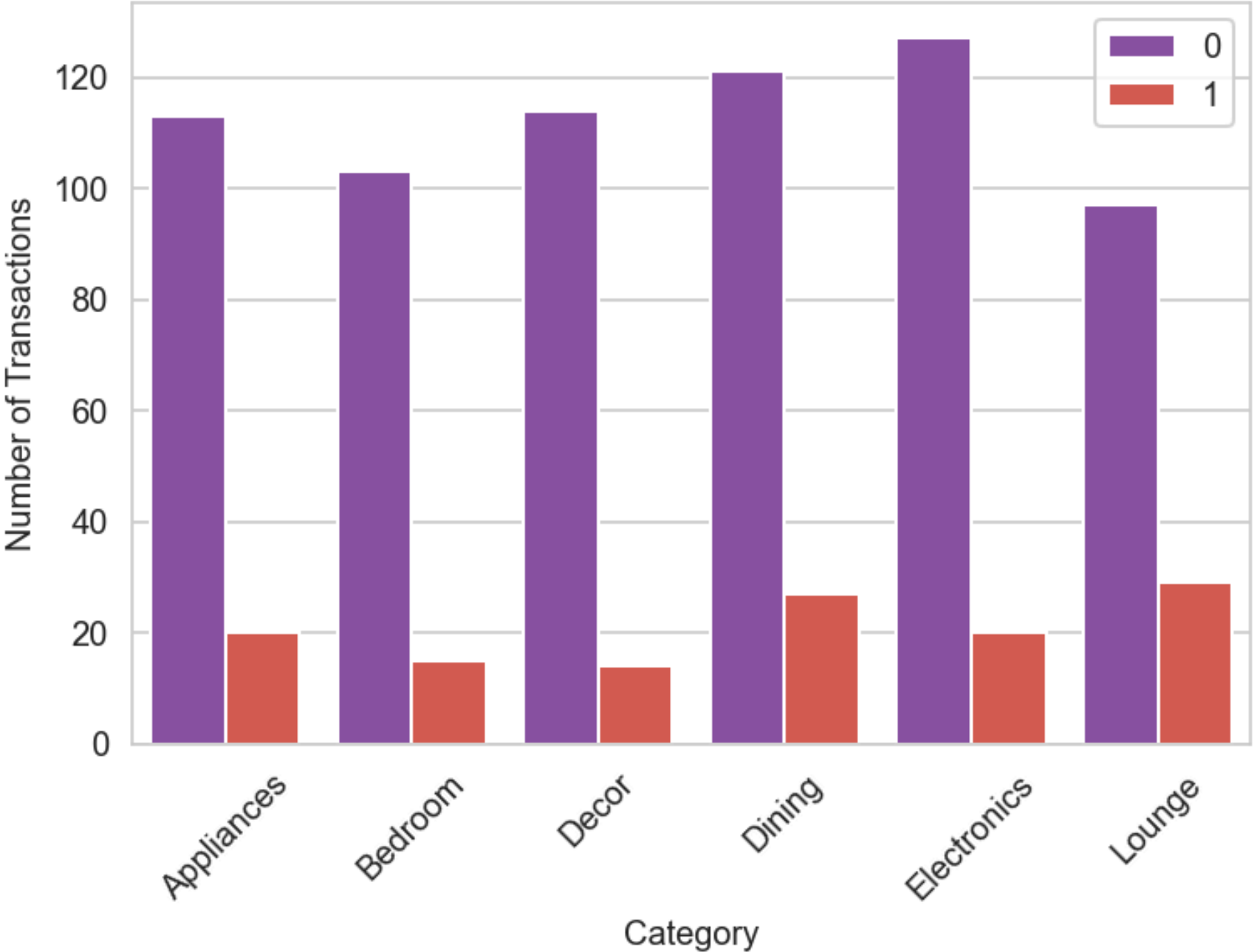
Satisfaction Distribution by Account Type



Total Satisfaction Score: 8401682
Cash : 3691692
Credit : 4709990

The average spent per transaction is of the top 10 customer ranges from R16 230.00 to R19 990.00. However, 60% of these customers churned.

Complaint(1) vs Non Complaint(0)



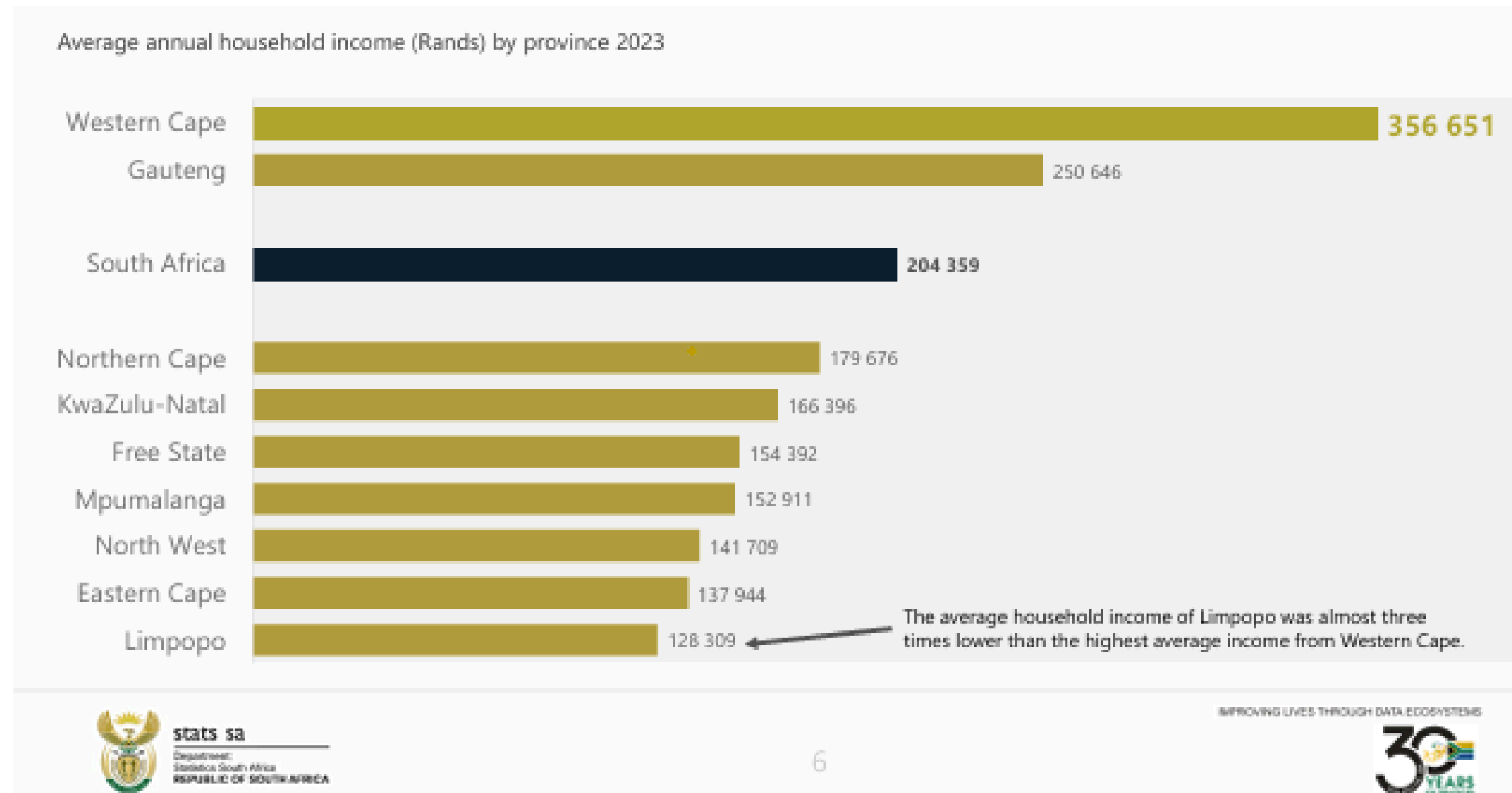
Regional Insights

Key factors:

- Loyalty score
- Average amount spent by customers in different regions
- Churn count, based on how many customers made only one purchase within 90 days

For this analysis, it is focused specifically on the following provinces: **Western Cape, Eastern Cape, Gauteng, Limpopo, and KwaZulu-Natal.**

Average household income(Rands)



Source : www.statssa.gov.za

Date accessed : 25 November 2025

Limpopo

Has the lowest average household income.

The average spent by customers in the region is:

Western Cape = R9 749.83

Eastern Cape = R10 274.11

📍 Limpopo = R10 327.91

KwaZulu-Natal = R10 882.43

Gauteng = R11 119.51

The Churn count per Region is :

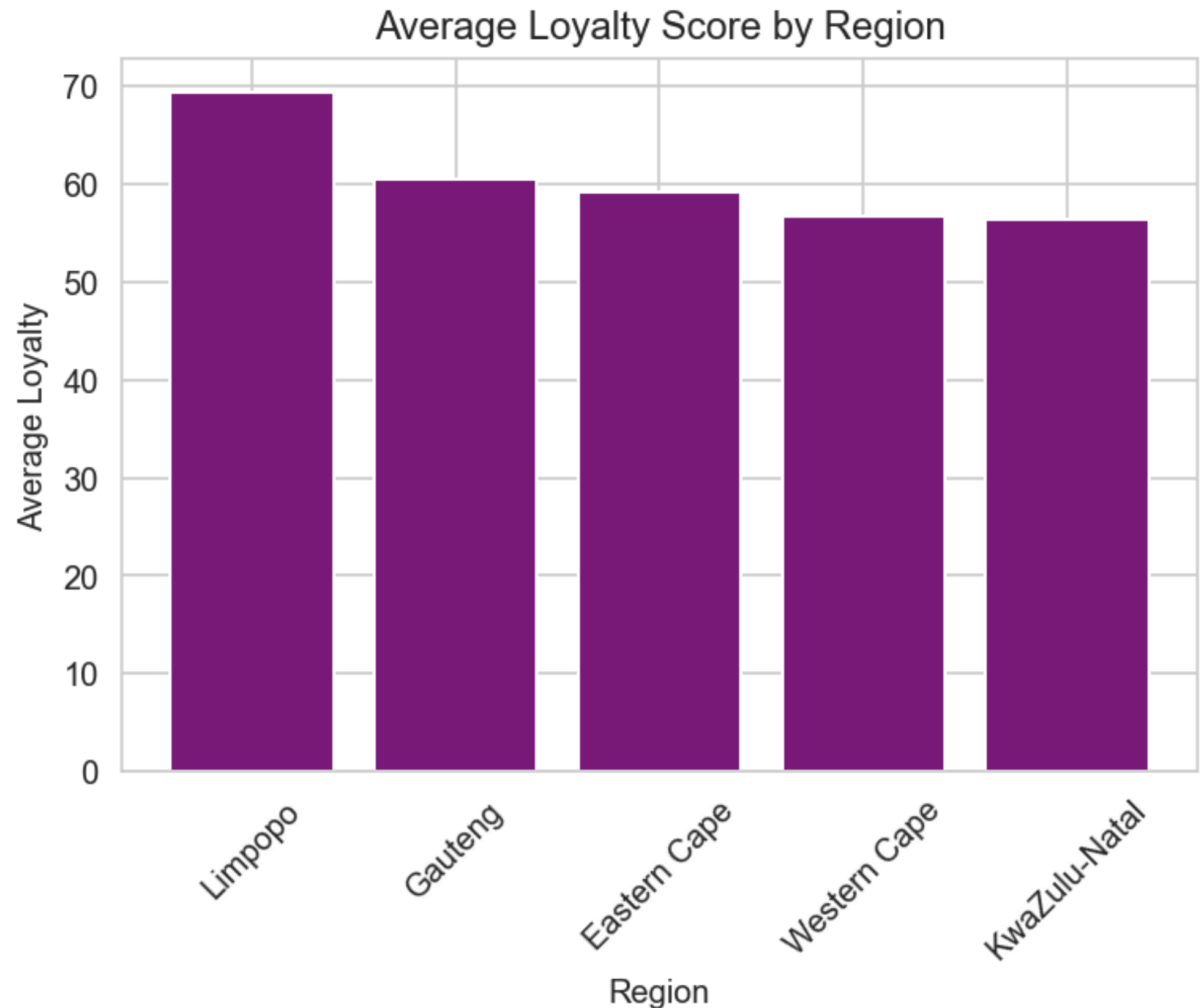
📍 Limpopo = 21

Western Cape = 41

Eastern Cape = 44

KwaZulu-Natal = 58

Gauteng = 87



Gauteng

Second highest average household income

The average spent by customers in the region is:

Western Cape = R9 749.83

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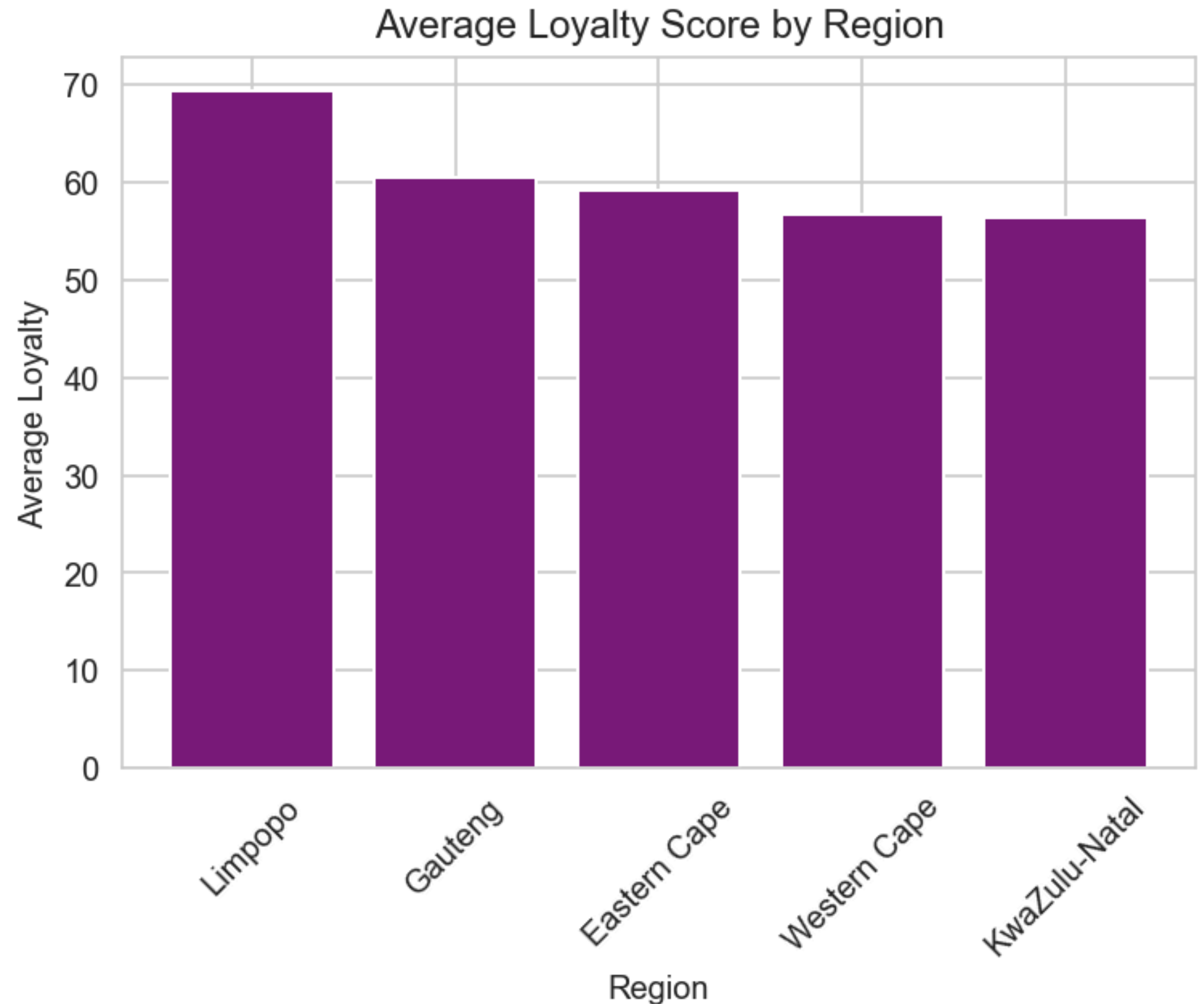
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Eastern Cape & Western Cape

Second lowest Vs Highest household income

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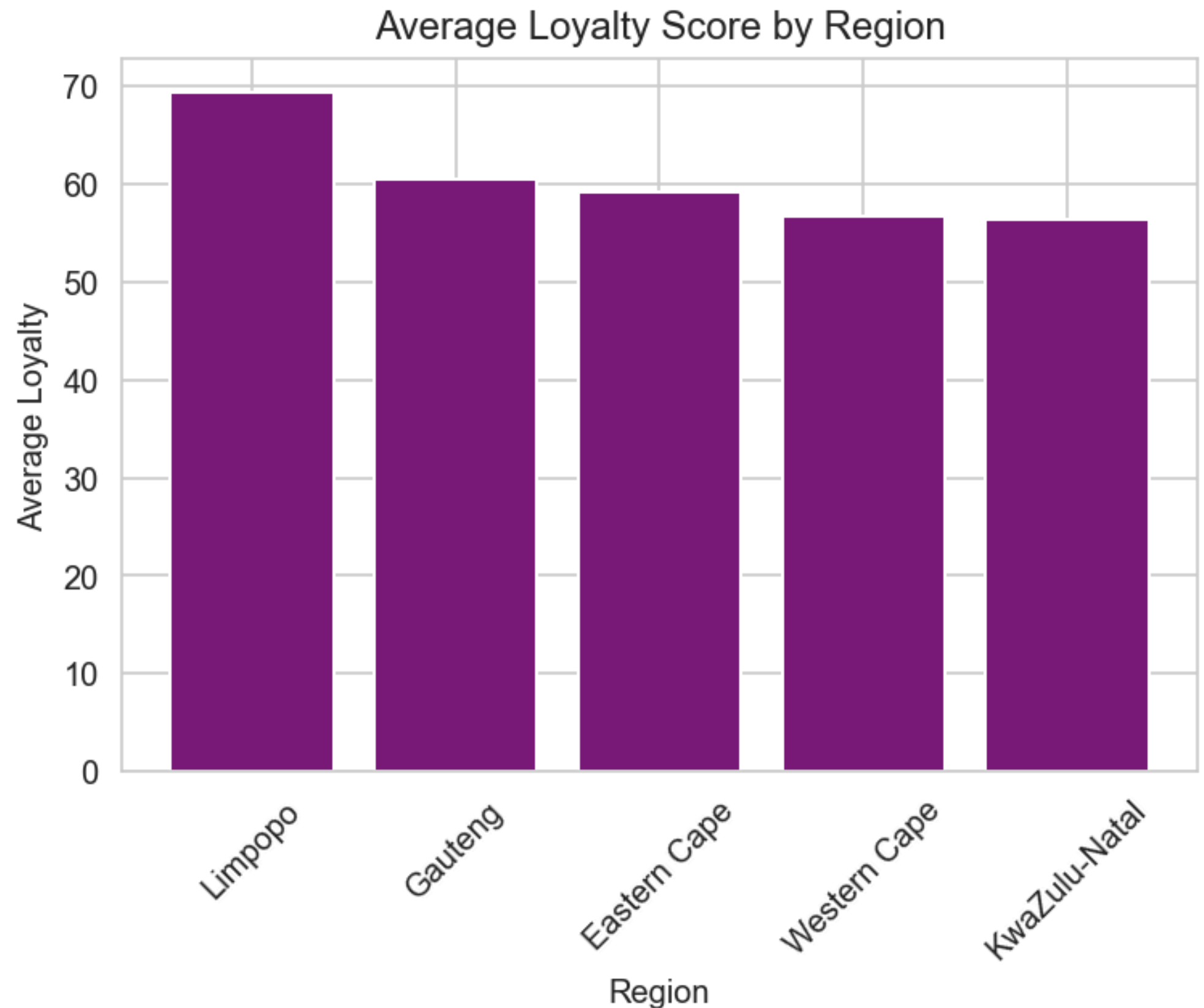
Limpopo = 21

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KwaZulu-Natal = 58

Gauteng = 87



KwaZulu-Natal

Just below the national average in household income

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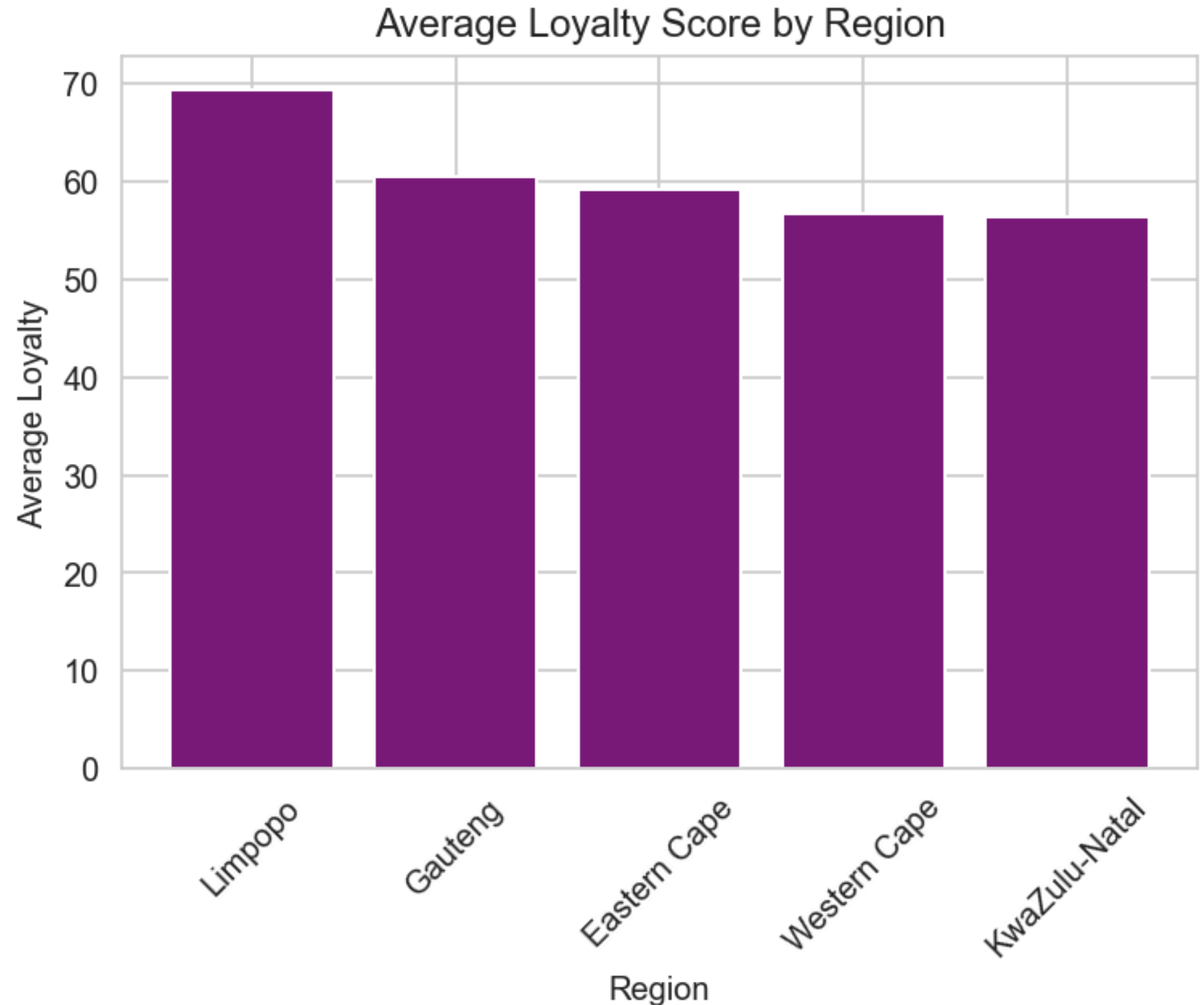
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Churn Prediction Model

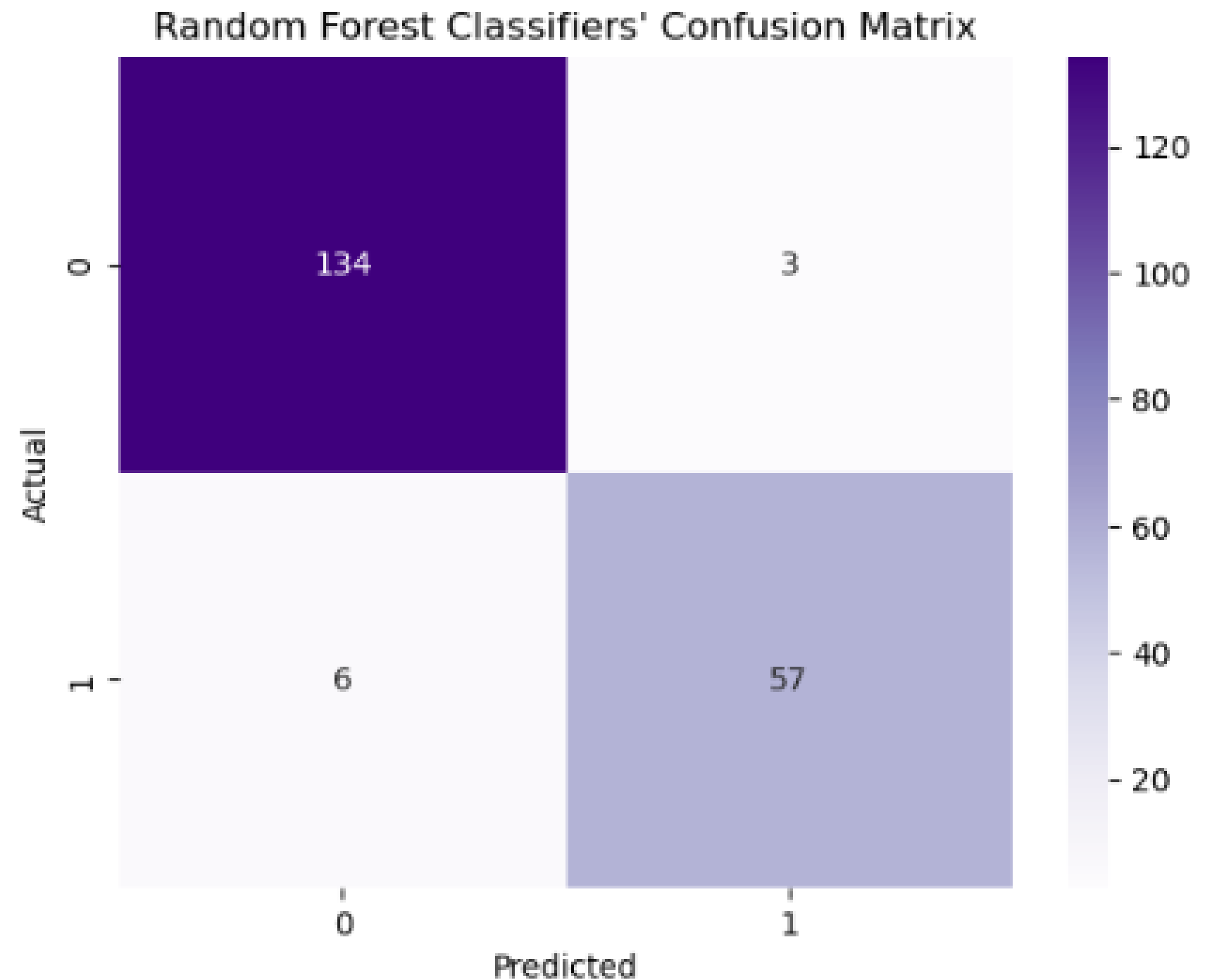
The algorithm used is called **Random Forest Classifier**.

The model's performance:

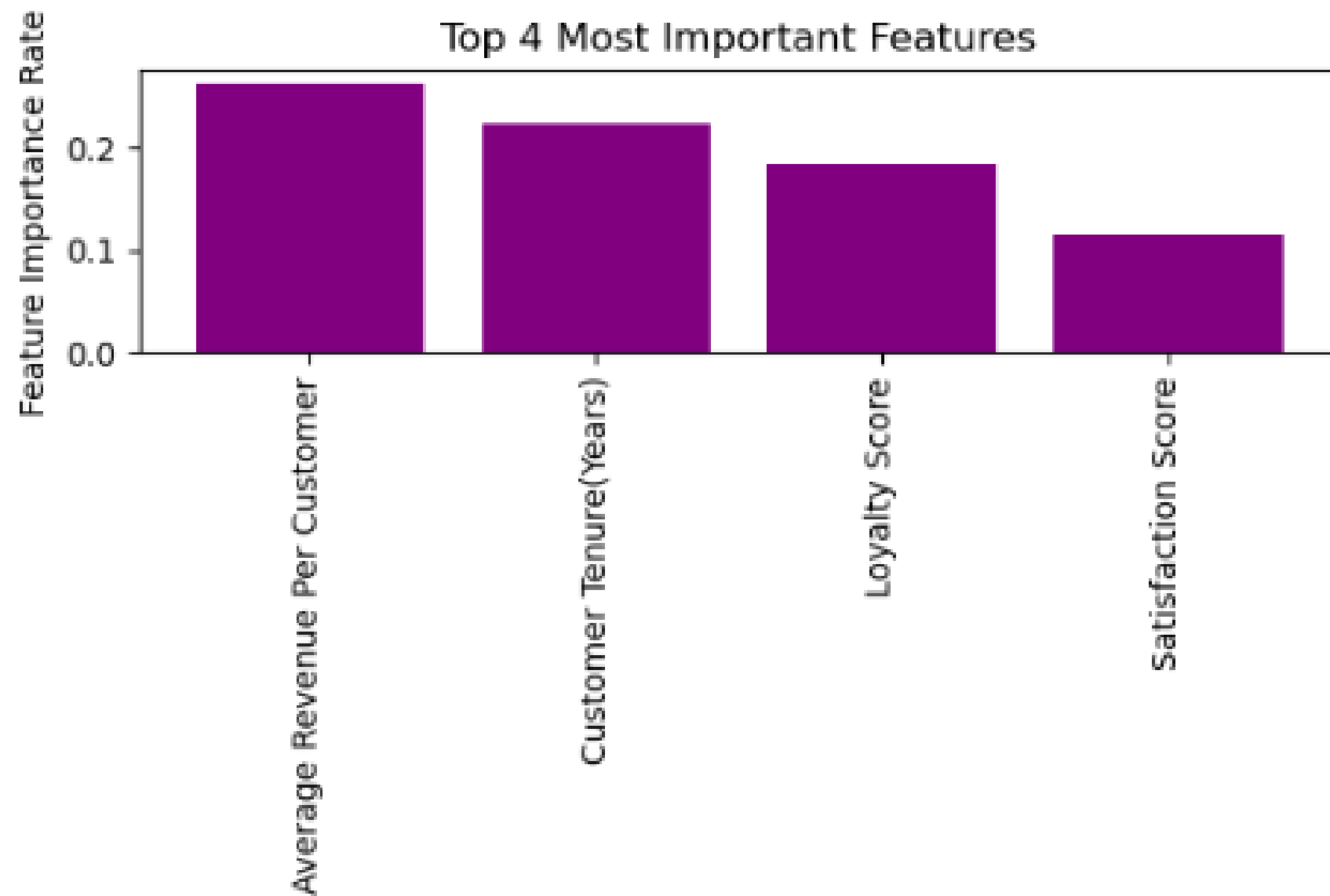
- Accuracy: 95%
- Recall: 90% – tells how well the model flags customers who are likely to leave.

The purpose of the model is to focus on flagging potential churners, even on new unseen data. With the data used to make the model, it predicted that:

- 134 customers left, and this was actually true
- 3 customers were predict to have left, but this was false
- 6 customers were predicted to stay, but they actually left
- 57 customers stayed, which was correctly predicted



Likely To Churn



1. Average Spend per Customer

- Low spenders often don't see value.
- High spenders expect very good service and may leave after a bad experience.

2. Customer Tenure

- New customers are the most likely to leave.
- We need strong onboarding and early relationship-building.

3. Loyalty Score

- Low loyalty is a clear sign a customer might leave.
- Shows the need for personalised communication and better brand connection.

4. Satisfaction Score

- Low satisfaction means the customer's needs are not being met.
- Points to issues with service quality, store experience, or expectations.

Data Driven Recommendations

Enhance Loyalty

- Personalised communication
- Rewards based on buying habits
- Discounts on repeat purchases

Improve New Customer Onboarding

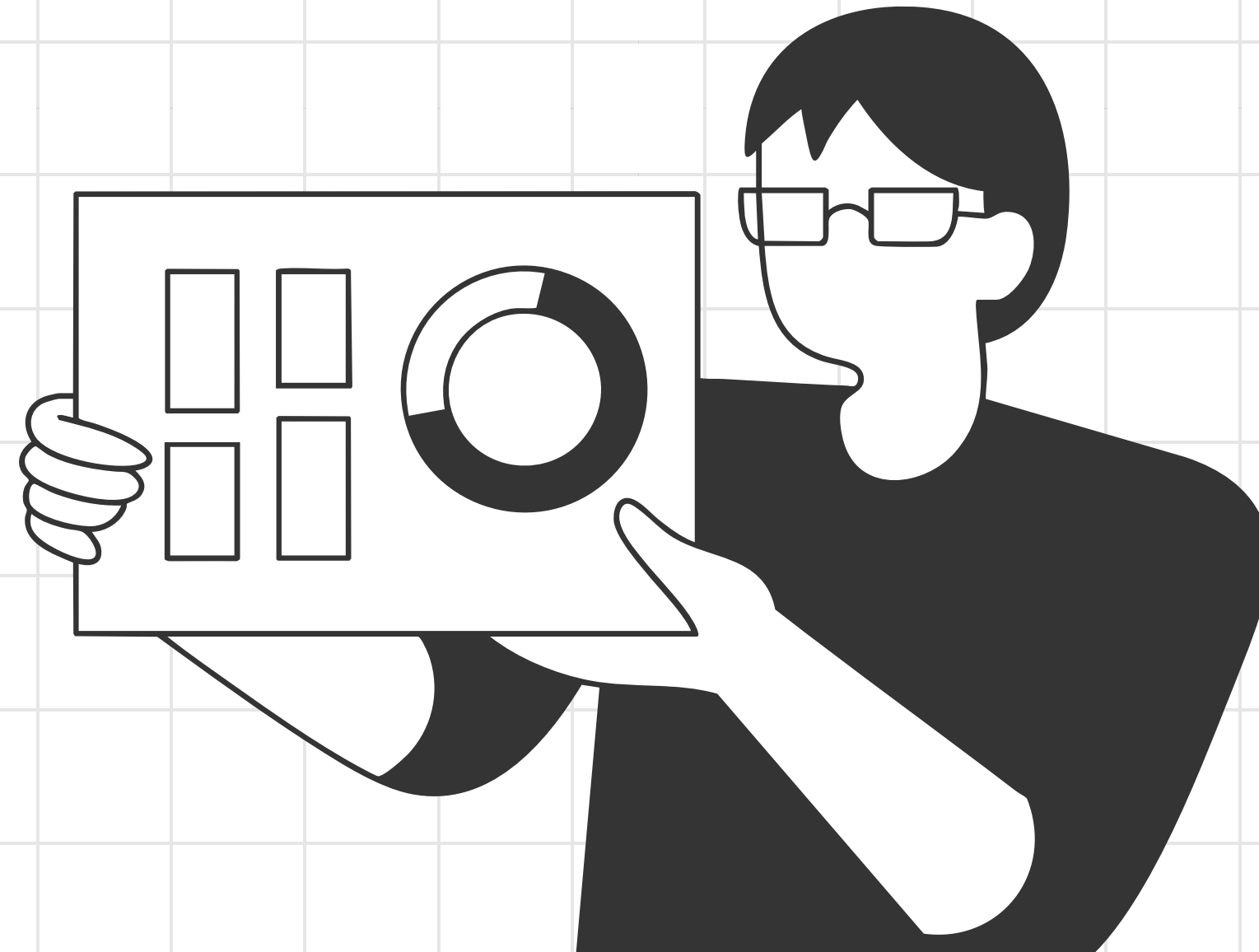
- Welcome deals (first 2–3 orders)
- Clear credit/payment terms
- Focus on high-value regions

Retain Long-Term Customers

- Strong customer service
- Follow-up communication
- Better delivery + clear assembly guides

Boost Customer Satisfaction

- Acknowledge issues
- Fix problems fast
- Offer practical solutions/compensation
- Train staff in empathy
- Learn from repeated issues





THANK YOU
ANY
QUESTIONS?

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